Homework 3 Summary

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Goal of the program:

The goal of this project was to expand my Farkle game that was developed in previous assignments into a complete playable game. The program allows players to roll dice, create scoring melds, bank points, reroll remaining dice, and continue playing until a target score is reached. For Homework 3, additional functionality was added for Full House score, multiplayer gameplay, score tracking between turns, and the official end game Farkle rules.

General Design Overview:

The project was designed using multiple classes, each with a specific responsibility.

- Farkle.java acts as the main driver of the program. It handles player setup, score tracking, turn rotation, and determining the winner.
- Turn.java manages an individual player's turn. It handles rolling dice, rerolling, banking points, and ending the turn.
- Hand.java stores the player's dice and manages moving dice between the unused dice and the meld.
- Meld.java stores the dice selected by the player and displays them.
- Dice.java represents an individual die and contains functionality for rolling and displaying ASCII dice.
- Score.java calculates the score of a meld and determines special combinations such as Straights, Three Pairs, Triples, and Full Houses.
- Player.java stores player names and cumulative scores throughout the game.

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Unit Test Description:

One important unit test was the Full House scoring test.

```
testScore("Full house", new int[]{2, 2, 4, 4, 4}, 1500);
```

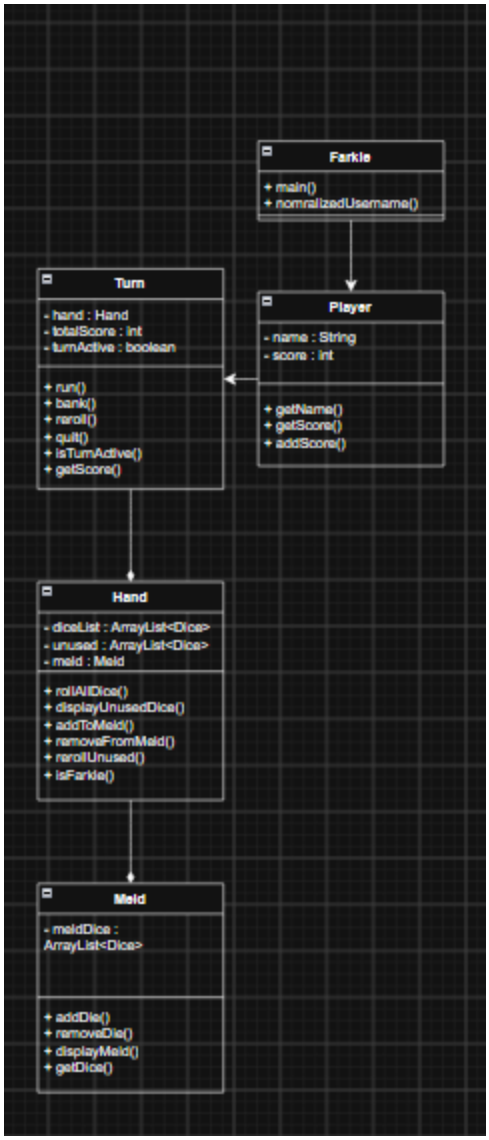
This test creates a meld containing a Full House and checks that the scoring system returns 1500 points. The test confirms that the Full House is detected properly and that the player does not receive additional points for the triple contained within the combination.

Major Design and Programming Issues:

The biggest challenge was implementing multiplayer gameplay. The previous versions only had single player, and were only created in mind of single player. The solution was to sit and think a while, relearn all of the previous code, and then go from there. A Player class was added which stored the names and scores. The game loop was also re-coded to rotate through all the players until one of them reached the target score. Additionally, the end game rules were difficult. All the remaining players get one final score after someone reaches the score goal, and getting that to work was a pain.

Retrospective:

If more time was available, I would improve the user interface and create automated tests for multiplayer gameplay and endgame behavior. Another improvement would be using a gui instead of the console input and output.


Farkle
+ main()
+ normalizedUsername()
Turn
- hand : Hand
- totalScore : int
- turnActive : boolean
+ run()
+ bank()
+ reroll()
+ quit()
+ isTurnActive()
+ getScore()
Player
- name : String
- score : int
+ getName()
+ getScore()
+ addScore()
Hand
- diceList : ArrayList<Dice>
- unused : ArrayList<Dice>
- meld : Meld
+ rollAllDice()
+ displayUnusedDice()
+ addToMeld()
+ removeFromMeld()
+ rerollUnused()
+ isFarkle()
Meld
- meldDice :
ArrayList<Dice>
+ addDie()
+ removeDie()
+ displayMeld()
+ getDice()

```
-----------------------------------------------
|           Zag Farkle by Aiden Munger         |
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-----------------------------------------------
How many points would you like to play in this game? [10000] 1000
How many players are going to be playing? [1] 4
What is the player #1's name? [Unknown Player #1] Aiden
What is the player #2's name? [Unknown Player #2]
What is the player #3's name? [Unknown Player #3] Bob
What is the player #4's name? [Unknown Player #4] Joe
Aiden, it is your turn! Rolling dice...
Starting hand:
+-------+ +-------+ +-------+ +-------+ +-------+ +-------+
| o   o | | o     | | o     | | o   o | | o   o | | o   o |
|   o   | |       | |   o   | |       | |   o   | |   o   |
| o   o | |     o | |     o | | o   o | | o   o | | o   o |
+-------+ +-------+ +-------+ +-------+ +-------+ +-------+
    A         B         C         D         E         F

=== Farkle Menu ===
A - Add die to meld
B - Remove die from meld
C - Show current meld
D - Show unused dice
E - Bank points
R - Reroll unused dice
Q - Quit
```

```
Choose an option: e
Banked 1000 points!
Turn comlete. Total score: 1000

Current total points:
Aiden 500
Unknown Player #2 1000
Bob 0
Joe 0


================================
Unknown Player #2 started the endgame!
All remaining players get ONE final turn.
================================


Final turn for Bob
Bob, it is your turn! Rolling dice...
Starting hand:
+-------+ +-------+ +-------+ +-------+ +-------+ +-------+
| o     | | o     | | o     | | o     | | o   o | | o   o |
|   o   | |       | |       | |       | | o   o | | o   o |
|     o | |     o | |     o | |     o | | o   o | | o   o |
+-------+ +-------+ +-------+ +-------+ +-------+ +-------+
    A         B         C         D         E         F

=== Farkle Menu ===
A - Add die to meld
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Q - Quit
Choose an option:
```

```
=== Farkle Menu ===
A - Add die to meld
B - Remove die from meld
C - Show current meld
D - Show unused dice
E - Bank points
R - Reroll unused dice
Q - Quit
Choose an option: e
Banked 150 points!
Turn comlete. Total score: 150
================================
Winner: Bob
Final Scores:

Aiden: 500
Unknown Player #2: 1000
Bob: 1500
Joe: 150
================================
PS C:\Users\aiden\Downloads\Fall 2026\224\AidenMungerHW2>
```

```
ScoreTest started.

Straight passed.
Three pairs passed.
Single one passed.
Three threes passed.
Triple ones passed.
Full house passed.
PASS: Farkle detection
PASS: Player name
PASS: Blank player name
PASS: Whitespace player name

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Tests passed: 10
Tests failed: 0
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```